

CAMPUSPULSE AI

Abstract

CampusPulse AI is an intelligent, AI-powered campus management and decision-support platform designed to transform a traditional college management system into a **smart, data-driven and real-time campus ecosystem**. The system integrates multiple campus activities and services, including **students, faculty, departments, department events, campus clubs, club events, projects and teams, academic monitoring, complaints, transportation, traffic, parking, canteen operations, waste management, and energy monitoring** into a unified platform.

The proposed system applies **Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, predictive analytics, anomaly detection, and recommendation techniques** to analyze campus data and generate useful insights. The **Academic Risk AI** module identifies students who may require academic support based on available academic indicators and assists in recommending appropriate interventions. The **AI Recommendation Engine** recommends relevant events, clubs, projects, teams, and career-related opportunities. The **Complaints AI** module uses NLP-based classification to categorize complaints and support appropriate routing and prioritization. Computer Vision can be used for **waste segregation and traffic/vehicle detection**, while predictive models support **canteen food-demand forecasting, transport/ETA prediction, parking occupancy prediction, and energy anomaly detection**.

The platform also provides a **real-time campus dashboard** containing campus activity analytics, AI-generated insights, alerts, live operational status, event information, and an interactive campus map. A **Campus AI Assistant** enables users to interact with campus information through natural-language queries. The system can use **Python-based ML/DL libraries and Django for application development**, with **MySQL** as the database where required. Free and publicly available datasets, APIs, and simulated campus data can be incorporated during development, while the architecture allows actual college data to be integrated later with appropriate authorization.

The primary objective of CampusPulse AI is to demonstrate how **Machine Learning can be applied to solve practical problems within a college environment** by converting campus data into predictions, recommendations, classifications, anomaly alerts, and actionable insights. Thus, the project serves as a unified **AI-powered smart campus platform** that can support students, faculty, departments, clubs, and administrators while providing a practical foundation for future real-world deployment.